

A COUNTEREXAMPLE TO CONJECTURE 3 OF FRADELIZI–MANUI–MEYER–NDIAYE

ABSTRACT. Fradelizi, Manui, Meyer, and Ndiaye conjectured that

$$|P_E K|_n \leq 2^n |P_E(K \cap \mathbb{R}_+^m)|_n$$

for every unconditional convex body $K \subset \mathbb{R}^m$ and every n -dimensional subspace $E \subset \mathbb{R}^m$. We give an exact counterexample for $(m, n) = (4, 2)$, with ratio $1399/336 > 4$. A Cartesian-product construction yields counterexamples for every $m \geq 4$ and $2 \leq n \leq m - 2$. Together with the positive cases established in the cited work and the elementary case $n = m$, this determines exactly the dimension pairs for which the inequality holds universally.

1. THE CONJECTURE AND RESULT

A convex body $K \subset \mathbb{R}^m$ is *unconditional* if it is invariant under all coordinate sign changes. Let P_E denote orthogonal projection onto a subspace E , and let $|\cdot|_n$ denote n -dimensional Lebesgue measure on E . Fradelizi, Manui, Meyer, and Ndiaye posed the following inequality as Conjecture 3 in [1]:

$$(1) \quad |P_E K|_n \leq 2^n |P_E(K \cap \mathbb{R}_+^m)|_n, \quad \mathbb{R}_+^m = [0, \infty)^m.$$

Theorem 1. *For every $m \geq 4$ and every $2 \leq n \leq m - 2$, there are an unconditional convex body $K \subset \mathbb{R}^m$ and an n -dimensional subspace $E \subset \mathbb{R}^m$ for which*

$$|P_E K|_n > 2^n |P_E(K \cap \mathbb{R}_+^m)|_n.$$

Consequently, for $1 \leq n \leq m$, inequality (1) holds for all such K and E if and only if $n \in \{1, m - 1, m\}$. In particular, Conjecture 3 of [1] is false.

2. THE COUNTEREXAMPLE

Set

$$(2) \quad K_0 = \{x \in \mathbb{R}^4 : 4|x_1| + 3|x_3| + 3|x_4| \leq 1, \quad 3|x_2| + 4|x_3| \leq 1\},$$

and let

$$b_1 = (3, -1, 1, 2), \quad b_2 = (-3, 1, 2, 0), \quad E_0 = \text{span}\{b_1, b_2\}.$$

The set K_0 is a full-dimensional unconditional convex body. Let B be the 4×2 matrix with columns b_1, b_2 , and define

$$\Phi : \mathbb{R}^4 \longrightarrow \mathbb{R}^2, \quad \Phi(x) = B^\top x.$$

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Since

$$G = B^T B = \begin{pmatrix} 15 & -8 \\ -8 & 14 \end{pmatrix}, \quad \det G = 146,$$

one has, for every compact convex set $A \subset \mathbb{R}^4$,

$$(3) \quad \text{area}(\Phi(A)) = \sqrt{146} |P_{E_0} A|_2.$$

Indeed, $\Phi = \Phi \circ P_{E_0}$; if $y = Bc \in E_0$, then the Euclidean area element is $\sqrt{\det G} dc$, whereas $\Phi(y) = Gc$ has area element $(\det G) dc$.

Write $K_+ = K_0 \cap \mathbb{R}_+^4$. The six inequalities defining K_+ have fifteen four-element active sets. Nine give feasible nonsingular intersections, namely

$$V_+ = \left\{ (0, 0, 0, 0), (0, 0, 0, \tfrac{1}{3}), (0, 0, \tfrac{1}{4}, 0), (0, 0, \tfrac{1}{4}, \tfrac{1}{12}), (0, \tfrac{1}{3}, 0, 0), \right. \\ \left. (0, \tfrac{1}{3}, 0, \tfrac{1}{3}), (\tfrac{1}{16}, 0, \tfrac{1}{4}, 0), (\tfrac{1}{4}, 0, 0, 0), (\tfrac{1}{4}, \tfrac{1}{3}, 0, 0) \right\}.$$

The other six choices consist of four singular systems and the infeasible intersections $(0, 0, \tfrac{1}{3}, 0)$ and $(0, -\tfrac{1}{9}, \tfrac{1}{3}, 0)$. Thus V_+ is the complete vertex set of K_+ . Unconditionality gives

$$(4) \quad K_0 = \text{conv}\{\varepsilon \odot v : v \in V_+, \varepsilon \in \{-1, 1\}^4\},$$

where $\odot$ denotes coordinatewise multiplication.

Set $Q_+ = \Phi(K_+)$ and $Q = \Phi(K_0)$. Applying Φ to the generators above gives the exact vertex descriptions

$$(5) \quad Q_+ = \text{conv}\left\{(-\tfrac{1}{3}, \tfrac{1}{3}), (\tfrac{3}{4}, -\tfrac{3}{4}), (\tfrac{2}{3}, 0), (\tfrac{5}{12}, \tfrac{1}{2}), (\tfrac{1}{4}, \tfrac{1}{2})\right\}, \\ Q = \text{conv}\left\{(-\tfrac{13}{12}, \tfrac{13}{12}), (-1, \tfrac{1}{3}), (-\tfrac{5}{12}, -\tfrac{1}{2}), (-\tfrac{1}{16}, -\tfrac{11}{16}), \right. \\ (6) \quad \left. (\tfrac{13}{12}, -\tfrac{13}{12}), (1, -\tfrac{1}{3}), (\tfrac{5}{12}, \tfrac{1}{2}), (\tfrac{1}{16}, \tfrac{11}{16})\right\}.$$

The points are listed counterclockwise. For a compact exact certificate of these hulls, the corresponding half-space descriptions are

$$(7) \quad Q_+ = \{(s, t) : s + t \geq 0, \quad 9s + t \leq 6, \quad 6s + 3t \leq 4, \\ 2t \leq 1, \quad -2s + 7t \leq 3\},$$

and

$$(8) \quad Q = \{(s, t) : |27s + 3t| \leq 26, \quad |30s + 21t| \leq 23, \\ |36s + 68t| \leq 49, \quad |19s + 55t| \leq 39\}.$$

Indeed, every projected generator satisfies the relevant inequalities, while the vertices of the two half-space intersections are exactly the points in (5) and (6). Thus the hull calculations require only exact rational arithmetic.

The shoelace formula gives

$$\text{area}(Q_+) = \frac{7}{12}, \quad \text{area}(Q) = \frac{1399}{576}.$$

Combining this with (3),

$$(9) \quad \frac{|P_{E_0} K_0|_2}{|P_{E_0} K_+|_2} = \frac{1399/576}{7/12} = \frac{1399}{336} = 4 + \frac{55}{336} > 4.$$

Hence (K_0, E_0) is a counterexample to (1) for $(m, n) = (4, 2)$.

3. PRODUCT EXTENSION

Proof of Theorem 1. Fix $m \geq 4$ and $2 \leq n \leq m - 2$, and use the orthogonal coordinate decomposition

$$\mathbb{R}^m = \mathbb{R}^4 \oplus \mathbb{R}^{n-2} \oplus \mathbb{R}^{m-n-2},$$

with zero-dimensional factors omitted. Define

$$K = K_0 \times [-1, 1]^{m-4}, \quad E = E_0 \oplus \mathbb{R}^{n-2} \oplus \{0\}^{m-n-2}.$$

Then K is unconditional and $\dim E = n$. Moreover,

$$\begin{aligned} P_E K &= P_{E_0} K_0 \times [-1, 1]^{n-2}, \\ P_E(K \cap \mathbb{R}_+^m) &= P_{E_0} K_+ \times [0, 1]^{n-2}. \end{aligned}$$

Therefore, by (9),

$$\frac{|P_E K|_n}{|P_E(K \cap \mathbb{R}_+^m)|_n} = 2^{n-2} \frac{1399}{336} > 2^{n-2} \cdot 4 = 2^n.$$

For the classification, the cases $n = m - 1$ and $n = 1$ are proved in [1, Propositions 3 and 4], respectively. If $n = m$, then $E = \mathbb{R}^m$ and unconditionality partitions K , up to null boundaries, into 2^m congruent orthant pieces, so (1) is an equality. These are all the remaining cases. $\square$

Scope. The body K_0 is invariant under all coordinate sign changes, which is precisely the hypothesis carried by (1); it is, however, neither an ℓ_q -ball nor invariant under coordinate permutations. Thus the example disproves Conjecture 3 as stated, but it does not address Conjecture 2 of [1], nor the variant of (1) in which K is required to be invariant under coordinate permutations as well as under sign changes.

REFERENCES

- [1] M. Fradelizi, A. Manui, M. Meyer, and C. S. Ndiaye, *L_p -Rogers–Shephard type inequalities for L_p -zonoids and symmetric bodies*, arXiv:2607.03582v1 [math.MG], 2026.